

Research paper

Synergy for climate actions in G7 countries: Unraveling the role of environmental policy stringency between technological innovation and CO₂ emission interplay with DOLS, FMOLS and MMQR approaches

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ARTICLE INFO

Keywords:

Environmental Policy Stringency

Technological Innovation

Financial Development

DOLS

FMOLS

MMQR

ABSTRACT

The significant and indestructible repercussions of global warming have an enormous effect on economies worldwide. Inevitably the preservation of the environment remains a crucial element in the pursuit of sustainable development goals. The primary aim of this study is to assess the moderating effect of environmental policy stringency (EPS) between technological innovation (INV) and CO₂ emission and direct association between financial development (FD), and financial globalization (FG) and environmental quality among the G7 countries. This analysis is conducted using a dataset covering the time period from 1990 to 2020, employing the Dynamic Ordinary least Square (DOLS), Fully Modified Ordinary Least Square (FMOLS), Canonical Cointegration Regression (CCR), and Method of Moment Quantile Regression (MMQR) estimation approaches. The results of DOLS, FMOLS, & CCR indicates that INV*EPS increase environmental quality by -0.21168 , -0.21257 and -0.21255 , respectively. Further, the outcomes of MMQR demonstrate that combine effect of INV & EPS contributes -0.2003 to -0.2264 across all quantiles from 20th to 80th. This indicates that technological innovation leads to enhanced environmental quality by mitigating CO₂ emissions and implementation of more stringent environmental rules enhance the positive effects of green technologies by employing eco-friendly technologies to achieve carbon neutrality target. In order to effectively attain sustainable development goals, the study suggests that in economies that exhibit a higher degree of innovation, strong financial system, the adoption and implementation of environmentally sustainable technologies have a significant influence on the mitigation of CO₂ emissions and the enhancement of environmental conditions. The study findings suggest few policies to policy-makers: enhancing the green finance structure and strengthening the design of environmental regulations to promote green innovation. Research and development (R&D) activities have the potential to significantly contribute to the reduction of CO₂ emissions through the effective application of scale effects and the implementation of efficient marketing tactics.

1. Introduction

Over the last century, global warming, climate change, and environmental degradation have plagued the world (Wu et al., 2022). These issues are primarily triggered by human activity and energy consumption, which releases hazardous pollutants. Catastrophe and the climate disaster require an integrated response to an unanticipated rise in global temperatures (Wang et al., 2021) caused by massive environmental degradation consequences (Du et al., 2023). Maintaining the current

climate paradigm would have serious repercussions. Policymakers and scholars are concerned about carbon emissions' unsustainable environmental impact. They urge for a carbon-free environment to reduce CO₂ emissions (Raihan, 2023; Khan et al., 2023). The need to limit the average rise in temperatures to below 2 °C is escalating, with attempts continuing to reduce it to 1.5 °C. To reach carbon neutrality by 2050, CO₂ emissions must be decreased by 45 % frequently. Sustainable and renewable energy sources could replace fossil fuels (Liu et al., 2023c). To combat global climate change, financial support is needed (Fatima

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<https://doi.org/10.1016/j.egy.2024.07.035>

Received 8 January 2024; Received in revised form 15 July 2024; Accepted 15 July 2024

Available online 23 July 2024

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et al., 2023a). However, countries must take major steps to reach carbon neutrality.

Technical innovation (INV) reduces CO₂ emissions and improves sustainability. The environmental effects of economic expansion and INV are intensively investigated in academia (Fatima et al., 2023b). Technologically advanced economies have implemented persistent regulations to mitigate industrialization's environmental repercussions for 20 years (Su et al., 2021; Amin et al., 2023; Lin and Ma, 2022). In developed economies, environmental INV is crucial to sustainability (Chen and Lee, 2020). These developments have helped enhance energy resource efficiency and integrate renewable energy into the energy mix. Green innovation and energy efficiency reduce CO₂ emissions and boost the manufacturing process. Efficient and suitable environmental policies encourage the development of eco-friendly technology. In addition to rigorous environmental regulations, the development of eco-friendly innovations is a crucial aspect in efficiently reducing the impact of CO₂ emissions on our environment. Technological innovation is a crucial factor in significantly improving the quality of the environment.

Therefore, by promoting environmentally sustainable technologies and discouraging ecologically inefficient and carbon-intensive practices, rigorous environmental regulations may mitigate CO₂ emissions (Yirong, 2022). Hence, EPS particularly climate taxes, and environmental regulations can change production and consumption behavior (Xie et al., 2023), emphasizing environmentally sustainable consumption and energy-efficient product manufacturing. Stringent environmental regulations are becoming more prevalent to a sustainable ecosystem and an effective strategy to reduce CO₂ emissions through innovative environmental technologies (Tiwari et al., 2023). The implementation of the regulation is just one of several measures to mitigate the negative effects of rising global temperature and climate change. Countries employ different environmental protection methods. The G7 carbon tax and ETS are effective instruments for reducing emissions (Boroza et al., 2023). Porter and Linde (1995) argue appropriate and sustainable regulation of the environment might promote green innovation. Thus, the enactment of stringent environmental policies has played a pivotal role in fostering the advancement of eco-friendly innovations and the development of a sustainable environment.

However, economic progress has caused environmental degradation, causing policy discussion challenging. As demonstrated by the SDGs, the international community has realized the need to change its economic development paradigm to restore environmental balance. The drivers of economic growth must be updated to shift the growth pattern. This would require strict environmental rules, innovative technologies, and an extensive financial conduit reform. Global warming remained a major discussion at the 47th G7 Summit in Cornwall. The 2021 Swiss Re Institute report started this debate. Due to their failure to confront climate change, the G7 nations' economies may decrease 8.5 %. The significance of employing innovative technologies and securing climate funding in tackling this issue has been widely recognized. Furthermore, the significance of the 47th G7 Summit is further demonstrated by its contextual relationship with the COP27 conference. Therefore, the implementation of stringent environmental policies to foster use of more sustainable technologies and mobilization of financial resources may be of utmost importance for environmental sustainability in G7 countries. The SDG accomplishment report 2021, highlights G7 countries' concerns to achieve SDGs 7 and 13. (Sachs et al., 2021). In light of these circumstances, it may be imperative to build a novel policy framework focused on sustainable development goals (SDGs) that incorporates role of environmental policies to encourage green innovation to mitigate CO₂ emission. The necessity for this shift of policy serves as the fundamental premise for the current analysis.

The current study selected G7 economies due to the following rationale: G7 economies are significantly enhancing climate action. Climate Transparency (2018) indicates that the G7 countries control over 60 % of global net wealth and approximately 50 % of world GDP.

Through significant manufacturing and commerce expansion, G7 economies are persistently growing. Excessive concentration on these economic activities harms the ecosystem. The US ranks second in global pollution, after China. Environmental issues have increased significantly. By diversifying their energy portfolios, the G7 nations aim to reduce CO₂ emissions (Murshed, 2020). Ahmad et al. (2021) estimate that these nations consume 25 % and emit 30 % of global energy and carbon dioxide. Furthermore, the G7 countries rely significantly on non-renewable energy sources. However, most G7 nations import non-renewable energy. According to the EIA (2022) report, Japan, Italy, and Germany import 96 %, 84 %, and 64 % of their primary energy, respectively. These facts highlight the G7's polluting fuel dilemma. Certain statistics illustrate why, despite their economic wealth, some nations have failed to protect their environments.

In implementing strict environmental regulations and green innovation toward a low-carbon economy, the world's largest economies' climate change policies and actions are significant. These economies have the potential to actively encourage and endorse the shift towards a global economy that is free from carbon emissions. The G7's future climate policy and commitment to carbon neutrality are currently uncertain. Therefore, it is necessary to analyze the elements that can significantly contribute to achieving carbon neutrality. Several economists have emphasized the importance of environmental policies and INV in attaining a carbon-neutral economy (Ike et al., 2020; Meng et al., 2022; Aziz et al., 2022). However, these conclusions fail to consider the interactive nature of environmental policy stringency in relation to both technical innovation and CO₂ emissions. Therefore, this study seeks to fill the prevailing literature gap by investigating the moderating role of strict environmental regulations on the relationship between technological innovation and CO₂ emissions in G7 countries. This study aims to provide specific answers to the following research questions:

- (i) How do strict environmental policies and green technological innovation effect CO₂ emissions in G7 economies?
- (ii) To what extent does the adoption of environmental policies moderate the association of technological innovation and CO₂ emissions in G7 countries?
- (iii) How does a robust financial sector assist in achieving the Carbon neutrality targets?
- (iv) Does Porter's Hypothesis remain applicable when stringent environmental regulations are enforced in G7 countries?

The rationale for investigating these particular questions arises from the pace of CO₂ emissions and environmental deterioration. The globe's leading countries and policymakers are dealing with the challenge of finding innovative approaches to achieve sustainable economic growth and improve the excellence of life while safeguarding environmental sustainability. However, the harmonization of a uniform policy framework is necessary to effectively implement stringent environmental regulations that promote the advancement of innovative technologies and financial development to achieve a sustainable environment. This change has the potential to influence the pattern of economic growth. Thus, the second objective of this study is to propose a policy & framework that aligns with SDGs 7 & 13 and addresses the negative environmental externalities caused by the economic growth patterns in the G7 countries. In light of SDG 7, there is a need for cost-effective, readily available, and environmentally friendly energy. Moreover, the increasing economic growth requires more energy consumption, resulting in increased emissions and posing a huge threat to both humanity and the environment. While the primary focus of this policy framework is on the G7 countries, it can also be applied to other advanced countries that intend to restructure their policies towards achieving environmental sustainability. Given its alignment with the goals of SDGs 7 and 13, this policy framework possesses the potential to serve as a reference point for countries seeking to adopt similar initiatives. It is appropriate to conclude that having a complete SDG paradigm

is crucial to accomplish the objectives of the 2030 agenda. Therefore, it is crucial to prioritize the contributing variables that assist policymakers in attaining the SDGs.

The present debate provides several insights concerning the environmental sustainability within the G7 countries: **First**, the G7 economies continue to face challenges in their pursuit of meeting the targets specified in SDGs 7 and 13. **Second**, the current renewable energy structure in G7 countries is inadequate in achieving environmental sustainability. Therefore, the G7 countries must modify their current energy structure by implementing more strict environmental rules and acquiring green technologies to increase the proportion of renewable energy sources to attain environmental sustainability. **Third**, considering the extensive efforts undertaken by G7 economies to promote environmentally friendly innovation and enforce stringent regulations for environmental sustainability, A significant aspect is to analyze the association between financial development, green technological advancement, stringent environmental regulations, and foreign direct investment to reduce CO₂ emissions in these nations.

Fourth, Despite the dearth of legally binding obligations, countries have adopted aggressive carbon reduction targets, which experts argue necessitate significant structural transformations in both economic and technological policies. The primary aim of this study is to assess the degree to which environmental policies are implemented, such as government policies or initiatives implemented by organizations, contribute to encouraging the advancement of green technological innovation in pursuit of carbon neutrality. These targets anticipate more strictness in environmental policies among G7 economies. This study contributes to the existing body of literature by employing extensive methodologies to empirically examine, the interactive relationship among environmental policy stringency, green innovation, and environmental quality by employing DOLS, FMOLS, CCR, and MMQR approaches. Furthermore, the application of advanced panel estimations, such as the MMQR, facilitates the examination of the heterogeneous impact of regressors across different data distributions. Further, we employed several econometric techniques in this study, including the DOLS, FMOLS, CCR, and Dumitrescu-Hurlin panel causality. To the best of our knowledge, the application of these techniques has not been prevalent in the field of environmental literature, particularly for the G7 countries. The study utilizes recently acquired data regarding the level of stringency in environmental policies and the extent of technical innovation related to the environment. This data primarily pertains to the G7 countries and covers the period from 1990 to 2020.

Finally, the initiation of this debate holds significance in alignment with the objectives established during the 47th G7 Summit, subsequently supplemented by the COP27 Summit and SDGs agenda. The current study aims to address the lack of academic research on the multipronged policy design focused on achieving environmental sustainability within the framework of SDGs 7 & 13. This study has the potential to make a valuable contribution to the existing body of literature in the field of energy economics. It additionally proposes a policy structure but also stimulates debate within the global environmental policy platform regarding potential strategies for realigning current policy regimes across nations to promote environmental sustainability. Furthermore, this study contributed to the existing body of knowledge by identifying numerous variables that influence environmental quality through the application of the Porter hypothesis framework. Porter and van (1995) suggested that effective environmental regulations can encourage industries to engage in technological innovation. Environmental regulation stimulates technical innovation and mitigates environmental degradation. Thus, it is significant that the selection of the G7 countries for this study is accomplished purposefully. This selection is made to initiate a discussion within the global environmental policy platform concerning the exercise of policy reorientation among the world's top economies. The results of this research will help policymakers in this region and other emerging nations in formulating suitable policies to gain a better understanding of the impact of these factors on

the degradation of the environment.

The rest of the work is organized as follows. **Section II** provides an exposition of pertinent literature studies and theories. **Section III** provides a comprehensive explanation and analysis of the data sample, variables, econometric technique, and the rationale for performing these analyses. **Section IV** presents the empirical results and interpretations of the analysis. **Section V** presents a detailed discussion of the obtained results. In **Section VI**, conclusions and policy implications are drawn based on empirical evidence.

2. Literature review

2.1. Environmental degradation and technological innovation nexus

The Paris climate-change agreement indicates a significant achievement in the global effort to address the challenges associated with global warming and foster the development of a low-carbon global economy. However, it is generally accepted that the majority of countries, including those classified as developed nations, will need to undertake substantial initiatives to fulfill their respective commitments (Horobet et al., 2021; Xing et al., 2023; Di et al., 2023). By considering the driven quality of the objectives, increased interests of the countries' result in limited dissents. The collaborations highlight the rising promise for funding in innovative technologies that are designed and adapted to tackle environmental challenges (Lyu et al., 2020; Kumar and Radulescu, 2024). The subsequent commitment outlined in the contract demonstrates a persistent devotion to executing appropriate initiatives intended for curbing climate change and global warming. The strategies can be enforced by customized measures that aim to deal with various situations. The target was set to reduce emissions, increase energy efficiency and consumption of renewable energy by 40 % by 2030 as per EU and UNSE4ALL agenda. This agenda highlights the crucial role of innovative technologies in mitigating environmental degradation at all levels. However, the combined effect of environmental policy stringency and innovative technologies on CO₂ remains inconclusive (Moridian et al., 2024; Sampene et al., 2021)

Green technological innovation (INV) implies the adoption of innovative industrial production techniques that emphasize environmental resilience and sustainability and promote effective utilization of resources (Song et al., 2019; Wang and Yang, 2021). INV is widely considered as vital resource for achieving environmental sustainability and overcoming persistent environmental challenges. Numerous studies have analyzed the impact of INV on environmental quality (Hu et al., 2021; Ullah et al., 2022; Fatima et al., 2023b). INV is considered as an effective policy measure to combat environmental degradation (Ikram et al., 2021; Mustofa et al., 2023). Innovations in green technologies have contributed to minimizing CO₂ emissions in developed countries (Aldieri et al., 2022; Aslam et al., 2023). The emergence of innovative technologies has contributed to improving effectiveness in various existing several initiatives, while also suggesting potential INV that significantly affect the environment. INV helps countries improve the effective use of renewable energy sources to mitigate environmental degradation and achieve net zero emissions targets.

2.2. Environmental degradation and environmental policy stringency nexus

The rising temperature and global warming and its consequent catastrophic ecological repercussions necessitate the implementation of strict environmental policies. EPS framework is particularly significant in these situations because it proactively encourages the adoption of green innovative technologies. These green practices are crucial to attain the zero-emission target of COP27 and Paris Agreement. EPS and INV are acknowledged as effective solutions for countries (Dmytrenko et al., 2024; Hassan et al., 2024; Yang et al., 2023; Patel et al., 2023). Both EPS and INV contribute to lower CO₂ emissions and stimulate green

growth. These measures are effective in curbing the climate change issue and thus improve environmental sustainability (Mehmood et al., 2024; Adebayo et al., 2023; Chegn et al., 2021). This discourse posits the importance of EPS and INV to tackle environmental issues. Further Cole et al., (2008) highlighted that energy consumption and industrial pollutions are positively associated with each other whereas, INV helps to lower CO₂ emissions. There are alternative perspectives among researchers who contend that carbon emissions are exogenous factors that necessitate additional constraints through environmental regulatory measures. According to Ringquist (1993), the enactment of robust regulatory mechanisms has played a crucial role in effectively attaining air pollution control objectives in multiple countries. According to Tol (2013), the predicted Pigou tax and its growth rate are insufficient in providing a justifiable basis for the environmental policy aims developed by the political establishment. The Porter Hypothesis posits that effective environmental rules have the capacity to foster innovation, leading to cost savings that may surpass the costs incurred for implementation (Porter and Linde, 1995).

Furthermore, the enactment of EPS, the development of sustainable innovations plays a vital role in effectively mitigating the adverse impact of carbon emissions on the environment. Developed economies have historically been responsible for the majority of carbon emissions (Sezgin et al., 2021). Thus, by reducing pollution and keeping emissions mitigation strategies implementable, these countries have enacted several policy initiatives to protect environmental sustainability. However, it is unclear how these policies will achieve the envisioned goals without appropriate perks and enticements (Ahmed et al., 2022). There are two distinct domains where the government can exert its effect. To motivate the use of green innovative technologies, it is crucial to minimize the cost of raw materials to maintain cost competitiveness of green goods and services (Nazir et al., 2023). Moreover, it is imperative to implement effective strategies that promote increasing public awareness and enact appropriate regulations that encourage the public to invest in ecologically sustainable goods (Khurshid et al., 2023). This suggests that implementing sufficient environmental regulations in the economy is a prerequisite to foster innovative technologies to attain environmental sustainability. Prokop et al., (2023) underscore the vital role of public policy for determining the need for green innovative technologies in the economy. Further, they also suggested that implementation of stringent environmental regulations generally stimulates the positive effects of green innovative technologies which are crucial in addressing CO₂ emissions challenges.

2.3. Environmental degradation and financial development nexus

Several research studies have been undertaken to analyze the effect of financial development (FD) on CO₂ emissions (Zhao and Yang, 2020; Khan and Ozturk, 2021; Hasni et al., 2023; Saud et al., 2023; Wang et al., 2024), following the introduction of EKC theory by Grossman & Krueger, (1995). From a particular perspective, FD plays a pivotal role in deriving economic growth. However, it is imperative to realize it also entails detrimental repercussions on environmental sustainability (Khan et al., 2018; Karliilar et al., 2023). Developed financial institutions offer credit alternatives to individuals and businesses at lower interest rates with fewer restrictions, thus leading to an increase in human activities and energy consumption which in turn triggers CO₂ emissions (Jinag et al., 2022; Kalemli-Ozcan et al., 2013). Contrarily, encouraging investments in innovative technologies and green practices to achieve environmental sustainability targets demonstrates the relevance and significance of FD. Investments in green sources, sustainable energy structures, eco-friendly innovative technologies, financial markets developments, and implementing strict environmental policies can be accomplished by giving sufficient access to funds for the promotion of environmental sustainability (Lahiani et al., 2021; Li and Wu, 2023). Moreover, FD has the potential to stimulate and expedite economic growth. FD encourages switch to renewable energy sources through

innovative activities, raising investments by pooling and deploying funds, encouraging FDI and streamlining the appropriate distribution of funds thus to achieve environmental sustainability (Mallick, 2020; Miao et al., 2022; Yu et al., 2024; Dong et al., 2024). The current emphasis on green finance has primarily highlighted the significance of FD in accomplishing the net zero emissions target and environmental sustainability (Patel and Mehta, 2023; Wang et al., 2024; Ozturk et al., 2024).

2.4. Environmental degradation and financial globalization nexus

The concept of financial globalization (FG) has become a subject of significant discussion, particularly in the integration of financial systems globally. Financing, investments, and businesses play a vital role in the development of economic liberalism (Murshed et al., 2021). Governments are constantly stimulating FDI, minimizing risks, expanding investment opportunities by implementing adaptive policies, and developing their economies. CO₂ emissions rise as a result of increased FG activities (Balsalobre-Lorente et al., 2023). FG provides more financial resources to invest in green projects in several sectors i.e, green energy, innovative technologies, and sustainable construction projects etc. (Nguea et al., 2023; Ramzan et al., 2023). The FG supported several international ecological preservation efforts to overcome environmental degradation challenges (Tovar García, 2012; Roper and Turner, 2020), and promote environmental sustainability (Haseeb et al., 2018; Liu et al., 2021; Sharif et al., 2022). From 1980–2016, G7 countries experienced twice a surge in the FG index from 52.01 to 80.20. UK has the leading experience of FG index of 88.87.17 (Qin et al., 2021). This surpasses the global average by a margin of 35 percent. This increase in FG asserts that FG might potentially exert a substantial influence on the environmental characteristics of these nations. Contrarily, according to the World Bank's 19th report, these countries together contribute to the overall global foreign direct investment (FDI) flows.

2.5. Environmental degradation and foreign direct investment nexus

Subsequently foreign direct investment (FDI) plays a significant role in facilitating FG, it is crucial to examine the environmental degradation associated with this phenomenon within the context of the Porter Hypothesis Framework (PHH). The discourse surrounding the precise relationship between FG and environmental issues remains unexplained within the existing body of academic literature. The study of Huo et al., (2022) analyze the reliability of the pollution haven concept and reveals that globalization amplifies the demands for self-regulation among businesses, even in countries with inadequate environmental protection regulations. Recent study of Sinha (2023) revealed that FG has a substantial and persistent influence on CO₂ emissions. This impact can manifest in either a symmetrical or asymmetrical conduct, contingent on the attributes of financial globalization. Leitao (2024) examined the association of human development, renewable energy and FDI with CO₂ emission in G7 economies by employing DOLS, FMOLS and ARDL panel model on the dataset from 1990 to 2019. Their findings revealed a negative association among FDI and CO₂ emission in G7 countries. Similarly, Borgi et al., (2024), examined the effect of eco-innovation, renewable energy, and FDI in G7 countries by employing regression models. They demonstrated a positive association between FDI and CO₂ in sample countries.

2.6. Research gap

In consideration of the increasing global concern regarding environmental pollution, prominent businesses, organizations, and researchers have demonstrated significant interest in exploring novel approaches to address this pressing issue. In order to achieve this objective, many researchers investigate the determinants that influence CO₂ emission. For instance, Ding et al., (2023) evaluate the direct

impact of disaggregated trade, technological advancements, income level, renewable energy sources, and energy demand on the consumption-based calculation of CO₂. However, the scope of their research renders their conclusions insufficient for other geographical areas. Similarly, a further study of Razzaq et al., (2023) focuses on the correlation between the environmental impact of foot-based consumption and the advancement of infrastructural development, along with technological innovation. The dearth of effective policy mechanisms is a significant factor contributing to the prominence of environmental issues. Consequently, there exists a dearth of empirical research examining the correlation between green technical innovation and CO₂ emissions among high-income nations that have implemented rigorous environmental policies. This study addresses a gap in the existing research by examining the role of environmental policy stringency as a moderator between technological innovation and CO₂ emission as well as direct relationship between financial development, financial globalization, foreign direct investment, and economic growth and CO₂ emission.

With the significant allocation of resources towards environmental protection, it has become of particular significance to evaluate the effectiveness of sustainability assessments and regulations. The present environmental strategy has the potential to significantly affect these outcomes, and this moderating effect is frequently perceived as an unnoticed or veiled impact. Without quantifying this implicit impact of environmental policy, policymakers may face challenges. It is necessary to align the existing policies in G7 countries to develop a reform program that is focused on achieving the Sustainable Development Goals (SDGs). The primary focus of this paper is to construct a reform agenda that is directed towards achieving SDGs 7 & 13. This agenda takes into account selected factors. This could be employed to describe the research's contribution. Furthermore, it is worth noting that previous research has predominantly relied on first-generation tests and basic econometric techniques. In contrast, our study has adopted innovative econometric methodologies such as FMOLS, DOLS, CCR, and MMQR, which have been recently introduced by Machado and Silva (2019). These advanced techniques address the issues posed by endogeneity and heterogeneity, enhancing the robustness and reliability of our findings. Also, the MMQR method provides results in different quantiles as well. Furthermore, our research has made a significant theoretical contribution by employing the Porter hypothesis model inside the framework of G7 countries

3. Data and material

Environmental policy stringency (EPS) and Green innovative technologies (INV) have a significant impact on the sustainability of the environment. To promote environmental sustainability, strict environmental policies are implemented to facilitate the optimal use of innovative technologies, adoption of green finance initiatives, stimulate financial development, encourage FDI. Therefore, the interplay between green technologies and environmental policies is crucial in enhancing environmental sustainability. In order to enhance the sustainable environmental quality within G7 countries, this study aims to assess the moderating effect of EPS on the relationship between INV and CO₂ emissions, and the direct influence of FG, FD, and EG on CO₂ emissions.

3.1. Data

This study employs data from G7 countries over a 30-year period from 1990 to 2020 to evaluate the moderating role of EPS between INV and CO₂ emission and direct role FD, FG, and EG with CO₂ emissions. The study period spans from 1990 to 2020, due to the availability of the data for the selected variables. The G7 countries have been selected for this study due to their notoriety for implementing rigorous environmental rules, maintaining robust financial systems, and aggressively promoting technology advancements in the context of climate change.

The data pertaining to certain variables is obtained from the databases of WDI and OECD statistics. Table 3.1 provides a comprehensive presentation of the detailed descriptions of variables.

The dependent variable selected to assess environmental quality is CO₂ emission, which is quantified as Metric Tonnes per Capita. The independent variables in this study are measured using specific indices. Environmental policy stringency is assessed using the EPS index, financial globalization is measured using the KOFFiGIdj index, economic growth is quantified using GDP growth (annual percentage), FD is evaluated using monetary sector credit to the private sector as a percentage of GDP, and FDI inflow is measured using net inflows as a percentage of GDP. INV is assessed using patent applications. The data pertaining to certain factors is obtained from many sources, including the OECD Statistics World Development Indicators (WDI), and the KOFFiGIdj Financial Globalization Index. Following is an explanation of each variable:

Carbon dioxide (CO₂) emissions are the primary greenhouse gas released as a consequence of human and economic activity. It contributes to environmental deterioration, specifically triggering global warming. Carbon emissions largely arise from the depletion of fossil fuels, including oil, coal, and natural gas, and cement manufacturing. The current study used CO₂ emissions as an indicator to quantify environmental impairment. The metric tons per capita of CO₂ emissions data is sourced from the World Bank database (www.data.worldbank.org). The scope of this includes the emissions of CO₂ resulting from the depletion of fossil fuels and the production of cement. However, it fails to include into consideration emissions from land use activities such as deforestation.

The **Environmental Policy Stringency (EPS)** index is a globally regulated composite index that evaluates various environmental policy measures, with a primary emphasis on policies related to climate change and pollution in the environment. The OECD (2021) established the EPS index (EPS) to measure environmental policies. An index with a range of 0 (which symbolize no stringency) to 6 (representing the highest degree of stringency). The stringency describes the level of strictness or intensity of environmental regulations that enforce penalties on activities that contribute pollution or degrade the environment. The present research used data for EPS index from the OECD database (OECD Statistics, www.stats.oecd.org).

Technological innovations (INV) are technologies that are designed to be advantageous to the environment. These technologies emit a lower GHG emission relative to existing technologies, hence exerting minimal strain on the environment. It primarily highlights modifying the goods and manufacturing facilities method to support sustainable development. Several indicators are used to assess environmental innovation, notably green financing, the percentage of spending on environmental

Table 3.1
Variables Description.

Variables	Description	Proxies	Sources
CO ₂ emission	Environmental Quality	Metric tons per Capita	WDI
INV	Technological Innovation	Patents application, residents	OECD Statistics
EPS	Environmental Policy Stringency	EPS Index	OECD Statistics
FD	Financial Development	Monetary sector credit to private sector (% GDP)	WDI
FG	Financial Globalization	KOFFiGIdj Index	KOFI
FDI	Foreign Direct Investment	FDI inflow	WDI
EG	Economic Growth	Gross Domestic Product (GDP Growth as annual %)	WDI

Note: Table 3.1 illustrates the selected study variables, a description of these variables with suitable adopted proxies. This table also demonstrates the sources of the data.

innovations, and the quantity of patents. The present research measures environmental technological innovation using patent data sourced from the OECD database (OECD Statistics, www.stats.oecd.org).

Financial Development (FD) is a crucial component of economic progress. Financial sector enables efficient transactions and reduces costs by offering various tools, associations, and structures. The measurement of financial sector development involves the use of many indicators that assess each significant aspect of this industry. FD is measured through monetary sector credit to private sector (% GDP). Therefore, the present research utilizes an indicator that has been produced by the International Monetary Fund (IMF). Data is collected from the IMF database (www.data.imf.org).

Financial Globalization (FG) is a comprehensive term that describes the growing linkages between countries through the movement of funds across national borders. Increased financial globalization is necessary to ensure that host economies benefit from financial flows, technological repercussions and experience environmental sustainability. Data is collected from KOF globalization Index (KOF Globalisation Index – KOF Swiss Economic Institute | ETH Zurich).

Foreign Direct Investment (FDI) inflows are expressed by the annually inflow of FDI as a percentage of GDP. The data for this proxy is obtained from the World Bank database (www.data.worldbank.org). FDI has a crucial role in fostering and stimulating economic growth. It promotes global economic integration and globalization by utilizing resource modelling, facilitating technological exchange, and facilitating financial mobility.

Economic growth (EG) is measured using the GDP indicator. Real GDP growth data, expressed as an annual percentage, is also obtained from the World Bank database (www.data.worldbank.org).

Employing interaction term of the stringency of environmental policy and key variable green technological innovation (INV*EPS) to analyze their influence on CO₂ emissions is an insightful strategy. Interaction terms provide a more thorough examination of how these factors cumulatively impact CO₂ emissions in relation to the strictness of environmental policies. Each of these variables is meticulously chosen to provide a thorough comprehension of several aspects that are essential to the study's emphasis on the strictness of eco-policy, innovative green technology, and financial development in the G7 economies. Their inclusion facilitates an in-depth evaluation of the interaction of environmental policy, technological innovation, and energy dynamics in influencing the formulation of sustainable development pathways towards achieving SDGs 7 and 13.

3.2. Model specification

The current research utilizes econometric estimations of panel data for various objectives. Panel data provide a larger sample size and improve the statistical accuracy to draw more significant conclusions (Hassan et al., 2020). Further, panel data reveals more apparent effects of the group of countries at various periods. Panel data provide strong statistics, more variations, lower multicollinearity, higher freedom, and more effective outcomes (Baltagi and Baltagi, 2008). Therefore, the subsequent econometric model intends to analyze the association between explanatory parameters and explained variable. The model for study is based in (Abbasi et al., 2024) and is as follows;

$$CO_{2it} = \theta_0 + \theta_1 FD_{it} + \theta_2 FDI_{it} + \theta_3 FG_{it} + \theta_4 EG_{it} + \theta_5 INV_{it} + \varepsilon_{it} \dots \quad (1)$$

$$CO_{2it} = \theta_0 + \theta_1 FD_{it} + \theta_2 FDI_{it} + \theta_3 FG_{it} + \theta_4 EG_{it} + \theta_5 EPS_{it} + \varepsilon_{it} \dots \quad (2)$$

$$CO_{2it} = \theta_0 + \theta_1 FD_{it} + \theta_2 FDI_{it} + \theta_3 FG_{it} + \theta_4 EG_{it} + \theta_5 INV * EPS_{it} + \varepsilon_{it} \dots \quad (3)$$

In the above equation, the variables "i" and "t" represent countries and time periods, respectively, whereas the term " θ " denotes the coefficients of parameters and " ε_{it} " is the error term. The dependent variable in this study is CO₂ emissions, while foreign direct investment

(FDI), financial globalization (FG), financial development (FD), economic growth (EG), and interaction of EPS*INV are employed as explanatory variables. The empirical estimating approach is illustrated visually in Fig. 1 and comprises seven steps that are explained in the subsequent sections.

3.3. Long term estimation techniques

The current study employed three cointegration techniques to assess the long run relationship among study variables. These techniques allow to evaluate the effect of regressors on regressand by selecting unbiased and competent estimators. Thus, these cointegration techniques include the DOLS (Dynamic Ordinary Least Square) by Pedroni (2000), Fully Modified Ordinary Least Square (FMOLS) and Canonical Cointegration Regression (CCR) by Park (1992).

Furthermore, the **DOLS technique**, developed by Stock and Watson (1993), integrates both lagged and leading values of the explanatory variable. This enables the error term in the symmetric co-integration equation to account for trends in the random regression. The DOLS method is useful for including cointegrated patterns that have parts integrated in a different sequence. This is accomplished by estimating the dependent variable based on the explanatory factors in levels, in addition to leading and lagging variables.

The **FMOLS method**, proposed by Phillips and Hansen (1990), is a semi-parametric approach to tackling association issues and is simultaneously unbiased. This approach is a modified version of least squares that takes into consideration the presence of endogeneity in the independent variables and the impacts of cointegration-related serial correlation. The FMOLS approach is useful for identifying spurious regressions by using ordinary least squares (OLS) regression methods on nonstationary data with unit roots.

The **CCR approach** (Park, 1992), particularly compared to FMOLS, provides a simple distribution and enables asymptotic chi-square validation. CCR involves the conversion of data using solely the stationary component of an integrating model. Following the completion of data processing, the cointegrating connection derived from the cointegrating model remains unmodified. The CCR modification in a cointegrating model diminishes the zero-frequency correlation between the error term and the regressors. The CCR technique generates estimators and chi-square tests that are simultaneously efficient and do not depend on extraneous parameters.

These three strategies are causally linked to both serial correlation and endogeneity.

These techniques, specifically FMOLS (non-parametric) and DOLS (parametric), employ distinct procedures. Considering their ability to effectively deal with endogeneity and serial correlation issues, these estimates exhibit a higher level of reliability in their long-run estimations. Furthermore, the DOLS operator demonstrates a high degree of accuracy in predicting time series data considering to its effective tackling of an issue of non-stationarity. Both Fully Modified Ordinary Least Squares (FMOLS) and Dynamic Ordinary Least Squares (DOLS) are expressed by the subsequent equation:

$$\Phi = \begin{bmatrix} a \\ \theta \end{bmatrix} = \left(\sum_{t=2}^T Z_t Z_t' \right)^{-1} \left(\sum_{t=2}^T Z_t y_t' - T \begin{bmatrix} \theta_0^{12} \end{bmatrix} \right) \dots \dots \quad (4)$$

The equation $Z_t = (X_t, D_t)$ represents the traditional version of the Fully Modified Ordinary Least Squares (FMOLS) method. The long run covariance matrix plays a significant role in the evaluation of the FMOLS methodology.

$$y_t = X_t \theta + D_{1t} \gamma_1 + \sum_{j=q}^{\gamma} \Delta X_{t+j} \gamma_j + v_{1t} \dots \quad (5)$$

Eq. (2) presents the traditional form of the Dynamic Ordinary Least Squares (DOLS) model, which incorporates the extension of

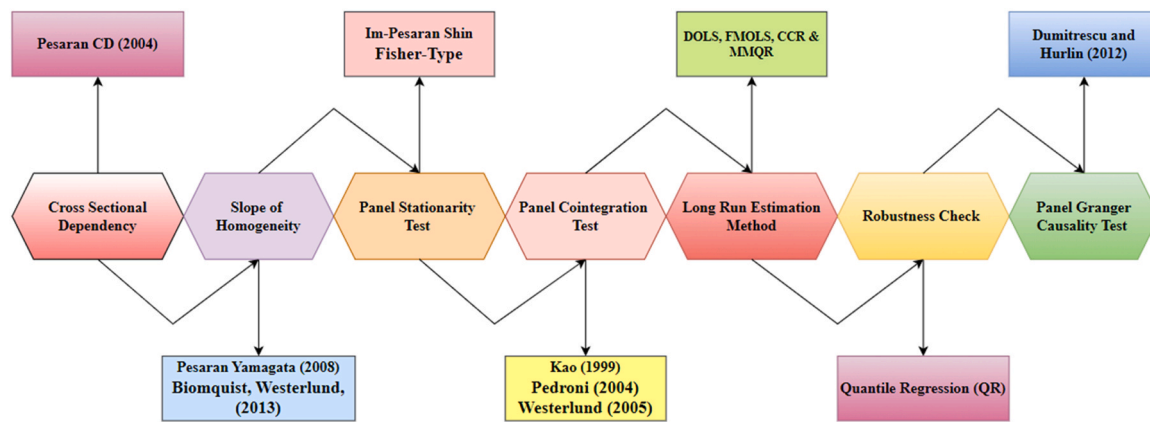


Fig. 1. Estimation Method Roadmap.

cointegration regression. This model takes into consideration the inclusion of lagged ΔX_t variables and leads related with the asymmetric error component in the cointegration equation. The DOLS estimator proposes that the potential long-term association between e_{1t} and e_{2t} can be identified by incorporating the integration of r leads and q lags of different regression coefficients.

The estimation approach for the CCR model relies solely on the use of regression analysis. However, the use of this method is both economically efficient and essential for tackling the linear regression aspect (Park and Zhao, 2010). Therefore, the identification of appropriate lags and lead orders is a significant challenge for the method being examined. In a broader context, the CCR estimators can be characterized by the following equation:

$$Y_t^* = \delta_{pq} Z_{pqt}^* + \mu_{pqt}^* \dots \quad (4)$$

The above equation represents the stationary transformation between Y_t^* and Z_{pqt}^* .

This study uses a unique estimating MMQR technique to investigate the association between the parameters for evaluation. The MMQR (Moment-based Quantile Regression) approach is especially well-suited for circumstances in which the model includes endogenous explanatory parameters, and the panel data exhibits individual-specific influences. Moreover, the MMQR exhibits its ability to produce reliable estimations in various conditions, particularly when the underlying model exhibits nonlinearity.

Moreover, it adeptly addresses the issues pertaining to heterogeneity and endogeneity. The MMQR demonstrates an inherent inclination to produce estimations that are distinctive across structural quantiles over time

The article provides an extensive examination of the estimation of conditional quantiles for the locational scale variation paradigm.

$$Y_{it} = \delta_i + W_{it}'\delta + (\gamma_i + X_{it}'\nu)U_{it} \dots \quad (6)$$

The equation provided involves a vector of parameters denoted as $(\delta, \delta', \gamma, \nu)'$. Additionally, the expression $P(\gamma_i * X_{it}' * \tau > 0)$ denotes a probability. The values (β_i, γ_i) where i ranges from 1 to n represent the fixed effects associated with each individual i . The matrix X is a vector of dimension k , composed of elements from the set W . These elements are distinct functions, with the l -th component being formally stated as:

$$X_l = X_l(W), l = 1, \dots, k$$

Eq. (5) suggests that the distribution and properties of the w_{it} element are consistent among individuals at a particular point in time. In a similar vein, the u_{it} component adheres to a symmetrical distribution, encompassing all individuals and time points, and has been defined to comply with the stipulations stated in Machado and Silva's (2019) research.

$$QY\left(\frac{\tau}{w_{it}}\right) = \delta_i + \gamma_i q(\tau) + W_{it}'\delta + X_{it}'\nu q(\tau)$$

In Equation 8, the vector X_{it} represents the IVs, including EPS*INV, FD, FDI, FG, and EG. On the other hand, $QY(\tau/X_{it})$ represents DV, i.e., CO₂ emission.

4. Empirical results

The proposed sequence for reporting the results is as follows: The initial step involves doing pre-estimation tests, which include a descriptive summary, a multicollinearity test, and a stationarity assessment of all variables using unit root tests. To examine the long run relationship between variables, this study employs DOLS, FMOLS cointegration regression tests. Further, MMQR approach is used to check the effect of study variables across different quantiles.

4.1. Descriptive statistics

Table 2 demonstrate the statistical summary of the variables for G7 economies. FD has the highest average value of 4.45208 with std. dev. Of 0.41242. While FDI has lowest average value of 0.0693873 with highest standard deviation of 1.443287 expressing the discrepancies of FDI in G7 economies.

4.2. Multicollinearity test

The Variance Inflation Factor (VIF) test is frequently employed to determine the presence of multicollinearity. Table 3 presents the outcomes of the VIF analysis, a statistical technique employed to assess the existence of multicollinearity across the examined variables. The VIF test findings suggest that the specified parameters do not exhibit multicollinearity.

The VIF values associated with the variables are less than 5, suggesting that there is no significant presence of multicollinearity among the parameters.

Table 2
Descriptive Statistics.

Variable	Mean	Std. Dev.	Min	Max
CO ₂	.3317839	.1613333	.0821374	.8124099
FD	4.45208	.4124207	3.26502	5.254419
FDI	.0693873	1.443287	−7.198535	2.54408
FG	4.374945	.1021583	4.043051	4.532599
EG	2.07735	.8248917	−1.975969	3.018951
INV	9.997347	1.321884	7.41637	12.5958
EPS	2.37129	1.051009	0.50000	4.89000
INV*EPS	3.040567	.5464392	1.367476	3.842103

Table 3
Multicollinearity Test.

Variable	VIF	1/VIF
FG	1.34	0.746219
FDI	1.26	0.790774
INV	1.47	0.681356
EPS	1.61	0.621800
FD	1.24	0.809547
EG	1.70	0.588792
Mean VIF	1.44	

4.3. Slope of homogeneity

This study performed a slope homogeneity test, considering the possibility that the slope variables determined by the regression estimation might vary among G7 countries due to their distinctive socio-economic features. The Pesaran and Yamagata (2008) and Blomquist and Westerlund (2013) offer an approach to determine the homogeneity of slopes in this specific situation. This approach entails the calculation of two test statistics (delta and Adjusted delta) to evaluate whether the slope coefficients are homogeneous under the null hypothesis.

The results reported in Table 4 support the presence of varying slope coefficients for G7 economies. Based on the statistical significance of both the delta and adjusted delta statistics, it reveals that there is heterogeneity in the slopes among G7 nations.

4.4. Unit root tests

In the framework of long-run panel estimating methodologies, it is imperative to ensure that all variables exhibit stationarity at level I(0) or first difference I(1). In order to verify the stationarity of data for long-term techniques, this study employs the Im-Pesaran-Shin and Fisher-type unit-root tests. Table 5 demonstrate that some variables, i.e. FD, FG and EG have unit root problem at level but become stationary at first difference in both tests for the G7 countries.

4.5. Cross sectional dependency

The CD test is a crucial diagnostic method before panel data analysis is performed. The influence of CD on estimates depends on a number of factors, including the type and extent of the CD test that is used. Furthermore, the current investigation uses of Pesaran's (2004) test, which verifies that the variables exhibit cross-sectional dependence (CD), as shown in Table 6.

The CD test results summarized in Table 6 indicate that the null hypothesis, which asserts that CD does not exist, is rejected at a significance level of 1 % demonstrating that each country is dependent on other country.

4.6. Panel cointegration tests

To determine the presence of a long-run association among the parameters under consideration, the present study employs the Kao (1999), Pedroni (2004), and Westerlund (2005) tests. The outcomes of the cointegration tests suggest that, at significance levels of 1 %, 5 %, and 10 %, the alternative hypothesis is confirmed, while the null hypothesis of "no cointegration" is rejected.

Table 4
Slope of Homogeneity.

	Delta	p-value
Pesaran, Yamagata. 2008		
Δ	10.640	0.000
Δ~Adjusted	12.798	0.000
Blomquist, Westerlund. 2013		
Δ	2.601	0.009
Δ~Adjusted	3.128	0.002

Table 5
Unit root Tests.

Variables	Im-Pesaran-Shin unit root test		Fisher-type unit-root test	
	Level	First Difference	Level	First Difference
CO ₂	−1.9988*	−7.8265***	−2.3818**	−11.7358***
FD	0.4149	−4.2534***	−0.0353	−5.881***
FDI	−4.5514**	−4.2534***	−6.1641**	−15.7912***
FG	−1.0730	−8.3179***	−1.1548*	−13.187***
EG	3.9333	−7.707***	3.0257	−11.9381***
INV	1.3409	−3.4835**	0.8117	−4.3965***
EPS	3.4507	−8.1590***	3.4380	−12.5119***
INV*EPS	2.3180	−8.550***	2.5222	−13.5901***

Table 6
Average Correlation Coefficient & Pesaran (2004) CD Test.

Variable	CD-test	p-value	corr	abs(corr)
CO ₂	23.63	0.000	0.988	0.988
FD	7.44	0.000	0.304	0.767
FDI	8.72	0.000	0.367	0.367
FG	13.29	0.000	0.555	0.571
EG	9.59	0.000	0.390	0.434
INV	11.35	0.000	0.462	0.726
EPS	22.43	0.000	0.914	0.914
EPS*INV	22.11	0.000	0.925	0.925

Note: Under the null hypothesis of $CD \sim N(0,1)$

and 10 %, the alternative hypothesis is confirmed, while the null hypothesis of "no cointegration" is rejected.

The results in Table 7 indicate that there exists a statistically significant and persistent association between the regressors and the regressands, even when accounting for the potential influence of CD. Consequently, we can confidently proceed to the next stage of estimation of parameters.

4.7. Long run estimation techniques

This study employs the DOLS, FMOLS, and CCR methodologies to investigate the potential long-term association between CO₂ emissions, foreign direct investment (FDI), financial globalization (FG), economic growth (EG), and interaction of technological innovation and environmental policy stringency (INV*EPS). Table 8 demonstrates the outcomes of the DOLS and FMOLS results.

The estimations obtained from the DOLS and FMOLS methodologies have provided empirical evidence supporting the assertion that EG has a significant and positive impact on CO₂ emissions in the long run due to the increase in human activities. Simultaneously, the remaining variables, primarily FD, FG, and FDI, EPS and INV, and their interaction INV*EPS have a detrimental effect on CO₂ emissions. Based on empirical evidence of DOLS, the findings indicate that a 1 % rise in the variables FD, FG, FDI, INV, EPS and the interaction of INV*EPS leads to a reduction in CO₂ emissions by −0.5012, −0.1491, −.25832, −.0659868,

Table 7
Cointegration Tests.

Kao Test	Statistic	p-value
MDFt	−9.7382	0.0000
DFt	−11.5264	0.0000
ADFt	−6.9396	0.0000
UMDFt	−19.6340	0.0000
ADFt	−13.3706	0.0000
Pedroni Test		
MPPt	1.5339	0.0794
PPt	−9.5143	0.0000
ADFt	−5.7156	0.0023
Westerlund Test		
Variance ratio	−2.2623	0.0118

Table 8
Cointegration regression Tests.

Variables	DOLS		FMOLS		CCR	
	Coef.	Std. Err.	Coef.	Std. Err.	Coef.	Std. Err.
FD	−0.05012***	0.017661	−0.05689***	0.00661	−0.05692***	0.006882
FDI	−0.01491**	0.00676	−0.01067***	0.00211	−0.01065***	0.002369
FG	−0.25832***	0.094307	−0.28879***	0.032675	−0.28898***	0.034357
EG	0.093589***	0.008941	0.091451***	0.003319	0.091464***	0.003442
INV	−.0659868**	.0196735	−.0589021***	.0153666	−.0587796***	.0152926
EPS	−.1112984***	.0081615	−.1152938***	.0062909	−.1156256***	.0064913
INV*EPS	−0.21168***	0.014044	−0.21257***	0.005205	−0.21255***	0.00524
_cons	2.140543***	0.433291	2.310454***	0.150235	2.311343***	0.157571

* denotes significance level at 10 %

** denotes significance level at 5 %

*** denotes significance level at 1 %.

−.1112984 and −0.21168, respectively. These findings are consistent with the earlier study of (Liu et al., 2023). Similarly, FMOLS results also demonstrate that there is a positive association between CO₂ emissions and EG, while all other variables are negatively associated with CO₂ emission this implies that when the values of these variables are augmented by 1 %, the corresponding CO₂ emissions will experience a decrease of −.5689, −.1067, −.28879, −.0589021, −.1152938, and −.21257 for FD, FDI, FG, INV, EPS and EPS*INV, respectively in FMOLS analysis. The outcomes of CCR demonstrates that FD, FDI, FG, INV, EPS and INV*EPS contribute to lower CO₂ by −5.692 %, −0.1065, −0.28898, −.1152938, −.1156256, and −0.21255 respectively. These results align with previous research of (Yang et al., 2023). The outcomes of DOLS, FMOLS and CCR reveals the long run association of selected variables. These results indicate that technological innovation has a significant role in driving the economy and enhancing competitiveness. Simultaneously, it maintains environmental sustainability, accomplishes development goals at lower costs by utilizing fewer energy units, and ultimately mitigates the detrimental impacts on the environment. Moreover, stringent environmental policies have been one of the most prominent tactics adopted to tackle the substantial threat to the quality of the environment and its sustainability over time. The objective is to eradicate unforeseen inefficiencies such as emissions in order to enhance environmental stability and foster sustainable development. The authors confirm the impact of stringent environmental regulations on green innovation. Moreover, stringent environmental restrictions have the potential to significantly promote the development of a green economy by encouraging technological innovation in industries that contribute to

pollution. Thus, coefficients of INV and EPS confirm the existence of Porter's hypothesis in G7 economies.

4.7.1. MMQR estimation results

Subsequently, we conducted an evaluation of the impact of FD, FDI, FG, EPS*INV, and EG on CO₂ emissions across each quantile. The current research employed the MMQR to examine and analyze the interplay of the variables. The outcomes of MMQR are presented in Table 9. A positive impact of EG on CO₂ is detected across all quantiles (0.20–0.80). Furthermore, the degree of the positive effect exhibits an upward trend as we progress towards higher quantiles. Consequently, within each quantile ranging from 0.20 to 0.80, an increase in EG leads to the stimulation of CO₂. The primary emphasis of this study, the GI (Technological Innovation), is significantly effective in reducing CO₂ emissions at all quantiles. The results demonstrated that an increase of one unit in INV efficiently reduces CO₂ emissions by 0.068210–0.50705 units. A reasonable explanation for this negative association is that INV promotes the advancement of eco-friendly technology. For example, by utilizing INV, the efficiency of resource development and consumption reduces the adverse environmental effects that arise from manufacturing and consumption activities, consequently culminating in a reduction in CO₂ emissions. Empirical evidence demonstrates that EPS has effectively addressed the rising issue of CO₂ emissions in G7 economies. More precisely, an increase of one unit in the strictness of regulations associated with the environment has led to a significant decrease in CO₂ emissions of 103564–1226800 units. The reason for this negative association is the strict environmental policies implemented by certain G7

Table 9
MMQR Estimation Results.

Variables	Quantiles			
	0.2	0.4	0.6	0.8
FD	−0.0300*** (0.0103)	−0.0424*** (0.0107)	−0.0578*** (0.0133)	−0.0875*** (0.0211)
FDI	−0.0115*** (0.0028)	−0.0111*** (0.0029)	0.0107*** (0.0035)	−0.0099*** (0.0056)
FG	−0.2859*** (0.0476)	−0.2857*** (0.0491)	−0.2853*** (0.0599)	−0.2850*** (0.0948)
EG	0.0912*** (0.0055)	0.0913*** (0.0056)	0.0918*** (0.0069)	0.0921*** (0.0109)
INV	−.068210*** (.005751)	−.064379*** (.005741)	−.058123*** (.006722)	−.050705*** (.008868)
EPS	−.103564*** (.005829)	−.108014*** (.005768)	−.115648*** (.006873)	−.1226800*** (.008666)
INV*EPS	−0.2003*** (0.0082)	−0.2059*** (0.0085)	−0.2129*** (0.0104)	−0.2264*** (0.0164)
_cons	2.0874*** (0.2181)	2.1843*** (0.2252)	2.3057*** (0.2755)	2.5394*** (0.4359)

** denotes significance level at 5 %

* denotes significance level at 10 %

*** denotes significance level at 1 %.

countries. These regulations discourage illegal manufacturing, and consumption, and make pollution or emissions activities costly. Consequently, these measures hinder countries from supporting non-renewable sources over sustainable ones. Furthermore, the INV and EPS outcomes in the current research align with those of Xie et al., (2023), who demonstrated a negative relationship between INV, EPS, and CO₂ emissions in the OECD countries from 1990 to 2020 by employing MMQR.

The results also reveal a consistently negative impact of FD on CO₂ across all quantiles ranging from 0.20 to 0.80. Furthermore, it has been shown that there is a consistent negative relationship between FDI and CO₂ emission across all quantiles ranging from 0.20 to 0.80. However, while moving towards the higher quantiles, the negative impact becomes less prominent. Therefore, the pollution-haven hypothesis is validated across all quantiles ranging from 0.20 to 0.80. Lastly interaction of INV*EPS demonstrates negative impact on CO₂ emission in all quantile (0.20–0.80), this indicates that 1 % surge in enactment of stringent environmental rules to foster the adoption of technological innovation contributes to reduce the CO₂ emission level by 0.2003–0.2264 % from lower to higher quantiles. Empirical evidence has conclusively illustrated that EPS fosters technological innovation to effectively address the growing issue of CO₂ emissions in selected G7 countries. The findings provide evidence for Porter's hypotheses. These findings are in line with the results of (Hassan et al., 2024). Their findings indicate that strict environmental regulations have a positive

influence on technological innovation, indicating that well-coordinated green regulations may serve as a win-win approach for promoting sustainability. Thus, it can be argued that the negative relationship is attributed to the adoption of green innovation and the implementation of stringent environment-related policies by G7 economies. These measures aim to hinder illegal production and consumption practices and to increase the costs associated with pollution and emissions activities. Consequently, countries refrain from preferring fossil fuels over sustainable energy sources.

Furthermore, Fig. 2 demonstrates the contribution of explanatory parameters to lower CO₂ emission across G7 countries. The graphs of EG show the, an increase in human and economic activities contributes to an increase in CO₂ levels across all quantiles. Overall, the graph of FD, FG, FDI, INV, and EPS contributes to lowering the CO₂ emission from quantile 20th to 80th. However, FG demonstrates that from quantile 40th to 60th, there is an increasing trend in the CO₂ emission, while FDI graph demonstrates an increasing trend in CO₂ emission from quantile 60–80 th. Furthermore, the combined effect of EPS*INV reveals a declining trend across all quantiles. The coefficients exhibit heterogeneity and yield a dynamic representation across all quantiles. Therefore, when conducting a comparative analysis of various panel estimators to offer a thorough elaboration of the relationship between variables, it becomes evident that the MMQR method emerges as a very effective and efficient strategy. Fig. 3 demonstrates the graphical outcomes of the variables obtained by employing econometric approaches.

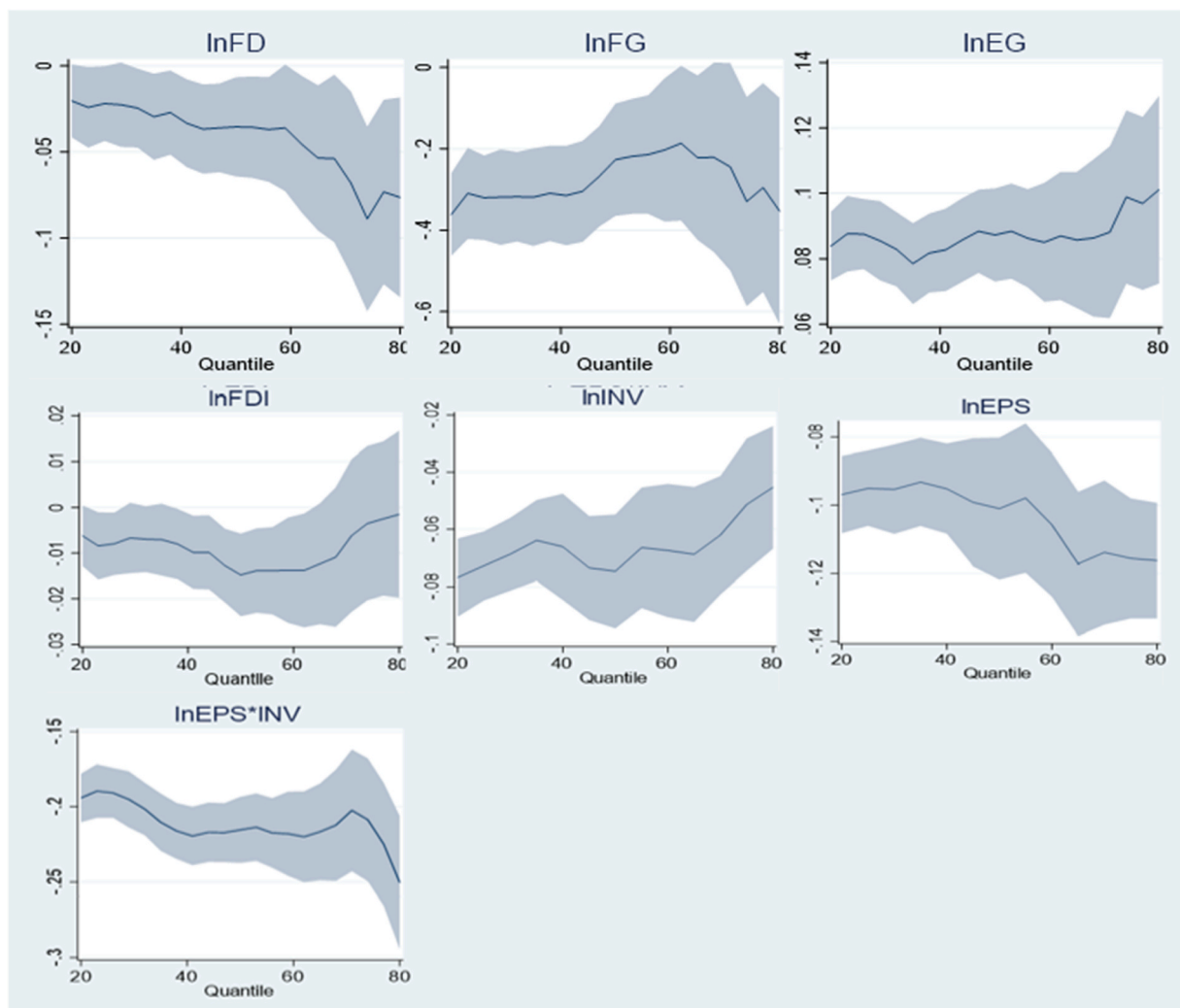


Fig. 2. Graphical representation of trends of explanatory variables contributing to mitigate CO₂ emission.

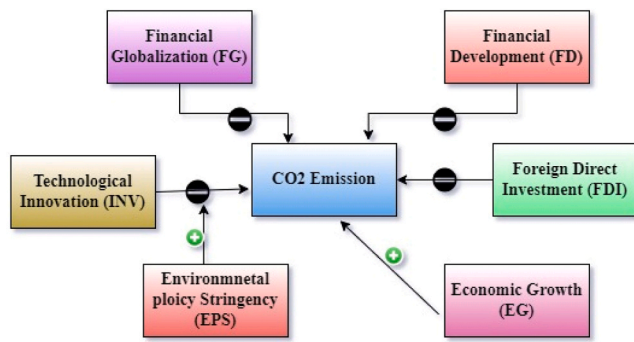


Fig. 3. Graphical representation of outcomes.

4.7.2. Robustness Check with Quantile Regression (QR)

Further, for the robustness of the results of MMQR approach, this study employed the Quantile regression (QR). The results shown in Table 10 confirm and support the findings of MMQR. The empirical findings of MMQR in Table 9 are supported by the robustness outcomes of FD, FG, FDI, EG, and EPS*INV.

This suggests that among the variables investigated in this research, EG in G7 countries stimulates CO₂ emissions. Conversely, FD, FG, FDI, and EPS*INV curtail CO₂ emissions. This suggests that the combination of green technology and strict environment regulations in G7 countries has the capacity to significantly decrease CO₂ emissions.

4.7.3. Panel causality test

Furthermore, Determining the causal relationship between fundamental variables is essential before formulating policy suggestions. The panel Granger causality test is used in this study to accomplish the stated research goals. The outcomes of this causality panel test are displayed in Table 11.

The results indicate that FD, FG and FDI, INV has unidirectional causal influence on CO₂ emission. This implies that policies related to FD, FG and FDI has unidirectional causality in G7 economies. Whereas, EG and EPS, and interaction of INV*EPS has bidirectional causality. This indicates that both regressors and regressand are interdependent which implies that changes in one variable can have significant impact on the other. This association emphasize the importance of considering EG and combined effect of INV and EPS in investment decisions and devising policies aimed at promoting environmental sustainability. These results demonstrate that these are factors are substantial for sustainable development with connected environmental aspects. In summary, results show that the factors influencing CO₂ emission have a beneficial impact on achieving SDGs.

5. Discussion of findings

Before discussing the implications, it is crucial to provide a concise

Table 10
Robustness Check- Quantile Regression.

VARIABLES	Quantiles			
	20	40	60	80
FD	-.020382*** (.010751)	-.0319714*** (.0131772)	-.0402978*** (.0201539)	-.0763461*** (.029431)
FG	-.3613353*** (.0515076)	-.312799*** (.0631294)	-.1959644*** (.0965537)	-.3530199*** (.1409986)
EG	.0838785*** (.0053121)	.0822653*** (.0065107)	.0863908*** (.0099579)	.1011431*** (.0145416)
FDI	-.0061923*** (.0033750)	-.0099374** (.0041363)	-.0138056** (.0063264)	-.0015279** (.0092385)
INV	-.0768023*** (.0069051)	-.0660632*** (.0093985)	-.0673607*** (.0117829)	-.0453265*** (.0108768)
EPS	-.0968644*** (.0057885)	-.0951528*** (.0067251)	-.1057886*** (.0107361)	-.1163198*** (.0086373)
INV*EPS	-.1942171*** (.0081775)	-.2208794*** (.0100227)	-.2198182*** (.0153292)	-.2504548*** (.0223855)
_cons	2.370866*** (.2364001)	2.322333*** (.2897397)	1.862846*** (.4431447)	2.842714*** (.6471297)

* denotes significance level at 10 %

** denotes significance level at 5 %

*** denotes significance level at 1 %.

Table 11

Dumitrescu and Hurlin (2012) Causality test.

Causality Directions	W-Stat	Z-Stat	Direction	Probability
CO ₂ ≈FD	0.5994	−0.7494	CO ₂ →FD	0.4536
FD≈CO ₂	2.5230*	2.8493		0.0044
CO ₂ ≈FG	0.8629	−0.2566	CO ₂ →FG	0.7975
FG≈CO ₂	2.9965**	3.7352		0.0002
CO ₂ ≈EG	2.1876*	2.2219	CO ₂ →EG	0.0263
EG≈CO ₂	2.2221*	2.2863		0.0222
CO ₂ ≈FDI	0.3869	−1.147	CO ₂ →FDI	0.2514
FDI≈CO ₂	3.1874***	4.0922		0.0000
CO ₂ ≈INV	1.5988	1.1203	CO ₂ →INV	0.2626
INV≈CO ₂	3.1409***	4.0052		0.0001
CO ₂ ≈EPS	2.9736***	3.6922	CO ₂ →EPS	0.0002
EPS≈CO ₂	3.7043***	5.0592		0.0000
CO ₂ ≈INV*EPS	3.0987***	3.9264	CO ₂ →INV*EPS	0.0001
INV*EPS≈CO ₂	3.7674***	5.1773		0.0000

overview of the main findings. Based on long-term estimations (DOLS, FMOLS, and CCR), it is evident that environmental regulations have a considerable impact on the relationship between technological innovation and CO₂ emissions, resulting in a beneficial contribution to environmental sustainability. This research aims to demonstrate how environmental rules govern the use of INV as a means to achieve the economic, environmental, and social goals of sustainable development. Green innovation is a significant catalyst for development and productivity. Simultaneously, it maintains environmental integrity, accomplishes development targets at a reduced expense by utilizing fewer energy units, and ultimately mitigates the detrimental impacts on the environment. The deployment of innovative technologies has the potential to optimize the efficiency of renewable energy sources and mitigate environmental degradation thus has the potential to achieve a net zero emissions target by 2050.

Further, it investigates the varying effects of these factors across different stages of economic development by employing the MMQR approach. Additionally, this technique has significant policy and administrative implications, along with suggesting potential possibilities for future research. When policymakers formulate environmental regulations that necessitate universal compliance from society, economic agents encounter a predicament. One perspective to consider is the necessity of prioritizing the preservation of the natural environment. Due to this rationale, government agencies and corporations are compelled to bear more expenses in implementing the requisite procedures to adhere to environmental regulations. According to Porter and Van der Linde (1995), it is argued that environmental regulations designed to be adequate and effective have the potential to encourage the development of green innovations. Further to the implementation of rigorous environmental policies, the development of sustainable innovations plays a crucial role in effectively mitigating the impact of carbon emissions on our ecosystem.

Therefore, this study extends upon previous research by employing

an empirical methodology to examine the effectiveness of stringent environmental regulations and technology advancements in attaining environmental sustainability goals. The findings demonstrate that the strictness of environmental policy and the association among green technological innovation substantiate that an increase in INV*EPS leads to a reduction in CO₂ emissions. This posits that EPS can facilitate the adoption of more environmentally friendly innovative technologies, leading to reduced CO₂ emissions. However, stringent regulations on innovative technology utilization in G7 nations may lead to a reduction in CO₂ emissions. Overall, stringent environmental policies can stimulate the innovation of greener and more efficient technologies that reduce CO₂ emissions. However, the impact of innovation (INV) on CO₂ emissions can be complex due to several factors such as the nature of the innovation, its implementation, and the broader economic and regulatory framework. For example, implementing carbon capture and storage technology or incorporating renewable energy technologies such as solar or wind energy can effectively decrease CO₂ emissions generated during electricity production. Technological developments of a similar nature have the potential to decrease energy use and the associated emissions in various sectors such as production, transportation, and energy-efficient structures (Abbasi et al., 2024). The impact of innovation in technology on CO₂ emissions, however, is not necessarily apparent. Technical developments, if not supported by appropriate regulations and incentives to ensure their sustainable utilization, can lead to increased emissions when they promote higher energy consumption (Ahmed, 2020a). However, this policy discourse also depends on the mobilization of financial resources to assist with this shift. Due to the government's inability to fulfill the \$100 billion financing of climate change commitment, the COP27 Summit expressed significant concern about the financial support for environmental projects (Azam, (2024).

The findings suggest that enforcing strict environmental policies encourages the adoption of green innovative technologies, which in turn contributes to improving environmental sustainability by reducing CO₂ emissions. Specifically, environmental policies have the potential to encourage the development of innovative solutions aimed at decreasing the environmental impact (Qin et al., 2021) associated with the consumption of resources such as materials and energy (Akram et al., 2023). These findings confirm the existence of Porter's hypothesis in G7 countries. This assertion posits that environmental policies strengthen the association between technological innovation and carbon emissions. Environmental regulations and incentives frequently encourage the use of cleaner technologies and practices such as energy efficiency, renewable energy, and reductions in emissions. However, an encouraging, attainable, and instantly implementable policy can play a role in transforming accessible and industrial conduct towards a sustainable economy and environment. Stringent environmental regulations can lead to increased costs for regulatory compliance, but the future benefits might compensate for these costs. The coefficients of INV*EPS confirm the Porter hypothesis, which states that Effective and appropriate environmental policies should promote the development of sustainable innovative technologies.

Furthermore, the empirical findings indicate that to achieve carbon neutrality, there exists an appropriate level of environmental legislation, green innovation, and research and development that would ultimately lead to a reduction in CO₂ emissions within the G7 economies. These study findings are aligned with the results of (Hassanpouryouzband et al., 2024; Sarpong et al., 2023). The analyzed the effect of environmental taxes, FDI, green technologies, trade openness, and structural modifications on environmental degradation by using MMQR in G7 countries. They suggested that environmental taxes, consumption of more renewable sources, green technologies, and modifications in existing energy systems contribute to lower environmental degradation, whereas, trade openness and EG have positive associations with CO₂ emissions. Based on this conclusion, it can be argued that strict environmental policies have the potential to curb the degradation challenge through the acquisition and adoption of green innovative technologies.

Furthermore, FD and CO₂ emissions are negatively related to each other. These findings demonstrate that mitigating environmental degradation can be achieved by deploying substantial financial resources to strengthen environmental regulations and policies to achieve sustainability. Economies with more developed financial systems exhibit more EG and improves environmental sustainability. These findings are consistent with the study of Abid et al., (2022). They suggested that a strong financial system contributes to lower environmental degradation by minimizing CO₂ emissions. They also concluded that renewable energy resources consumption and green innovative technologies help to reduce CO₂ emissions. From these findings, it can be argued that with a robust financial system, appropriate allocation of financial resources for green practices helps economies to achieve net zero emission targets, and SDGs 7 & 13. The negative association of FD and CO₂ emissions suggests that policymakers should implement policies and measures that foster development of financial system as well as long-run environmental sustainability

This research contends that accomplishing a zero-carbon economy requires the adoption and execution of appropriate policies that efficiently tackle environmental degradation problems and encourage green innovation to promote sustainable economic development. Several research studies (Liu et al., 2023; Sharif et al., 2022; Yu et al., 2022) demonstrates that environmental sustainability is positively correlated with a reduction in CO₂ emissions. Therefore, it is imperative to focus on promoting industrial and economic development while utilizing environmentally sustainable resources. The considerable impact of environmental policy plays a crucial role in fostering a sustainable environment through acquisition of green technologies. Consequently, implementing these environmental policies is vital in enabling G7 economies to attain their goal of carbon neutrality. The impact of green innovation on carbon emissions is primarily negative, thus necessitating the G7 region to prioritize the advancement of research, development, and green innovation as an approach to attain the objective of carbon neutrality. Finally, it is evident that EG significantly contributes to the increase CO₂ emissions. Therefore, the stabilization of financial sectors has the potential to foster a sustainable environment. The outcome will support G7 nations in attaining their objectives for SDG 7 and 13.

Moreover, G7 economies are implementing environmental measures and intensifying collaborations to tackle environmental catastrophes and expedite switching towards renewable energy sources, and adoption of innovative technologies globally to achieve net zero emissions by 2050. Through globalization initiatives, G7 countries strive to render these sustainable solutions the most intriguing alternatives globally (Zheng et al., 2023). Further, G7 economies seek to set a precedent by encouraging sustainable supply chains through public procurement. These economies support their persistent commitment to the Paris Agreement and SDGs agenda while encouraging private-public investments in future industries. They also recognize the necessity for further collaborations to overcome a deficit in shifting to renewable energy sources and reduce the costs of this change globally. G7 countries continue to provide support for R&D activities, and deployment of innovative technologies, recognizing their vital role in facilitating rapid shift to achieve net zero emissions target. In these aspects, G7 countries have made progress towards attaining net zero emissions targets. The premise is that through the implementation of strict environmental regulations, stimulating financial development, fostering green innovations, and switching to renewable energy sources can be expedited and a target of net zero emissions can be achieved in the future.

6. Conclusion

Policy makers and academia are investigating potential alternatives for sustainable economic growth and environmental preservation due to the enormous environmental degradation and depleting resources of fossil fuels. However, it is imperative to conduct further research and implement policies that promote INV and environmental sustainability,

particularly in highly developed countries such as G7 countries. In this regard, the present study examined the direct association of FD, FG, FDI, EG and CO₂ emissions. Further the study also analyzed the interactive role of EPS on the association of INV and CO₂ emission to achieve aim for carbon neutrality for G7 economies by employing DOLS, FMOLS, CCR and MMQR approaches for the period of 1990–2020. The results of unit root tests, IPS and Fisher type indicates that the study variables are stationary at level I(0) and first difference I(1). Further cointegration tests Kao (1999), Pedroni (2004) and Westerlund (2005) reveals the long run association of the variables that conform to the adoption of DOLS, FMOLS, CCR and MMQR approaches. The results of all these techniques demonstrates that FD, FG, FDI and INV*EPS contributes to lower CO₂ emission, whereas, EG increases environmental degradation due to increase in human activities in G7 countries. These findings posit that the technological innovation contributions to attain sustainable development continue to be a subject of discussion in strategy, management, and research realms. In order to contribute to ongoing discussions, the objective of this investigation is to evaluate the influence of INV on environmental sustainability, with the moderating effect of EPS. When examining the influence of EPS, INV, and other variables, it is important to consider their respective roles. The results indicate that the goal of carbon neutrality is feasible to attain within the G7 economies through the implementation or improvement of environmental regulations, the encouragement of green innovation, development of financial sector and the expedited progress of R&D in energy sector. These research results certainly make a significant contribution to the current sustainability discussion and also highlight the complex interaction between energy policy, green technological innovations, and energy consumption in influencing the environmental effects in G7 countries.

Following the announcement of an environmental catastrophe, a United Nations summit on Climate Change in Egypt (COP27) yielded alignment among countries to participate in collaborative and cooperative efforts to reduce emissions by 45 %, shift towards sustainable energy sources through the utilization of green innovation, and reinforce environmental regulations to attain these targets. In order to tackle the global crisis of climate change and the associated increase in global temperatures, researchers have conducted a considerable amount of research on the challenge of CO₂ emissions, emphasizing its pivotal role in addressing this global issue. However, only a limited number of researchers have incorporated it into their empirical models. Moreover, the existing scholarly literature lacks sufficient empirical evidence to elucidate the moderating effect of stringent environmental regulations on green technology and CO₂ emissions among the G7 countries by employing DOLS, FMOLS and MMQR techniques.

According to the empirical findings of this study, it is recommended that with the formulation of regional emission targets, adequate consideration should also be given to the establishment of consumption-based CO₂ emissions, an aspect that is often disregarded. The CO₂ approach illustrates the consumption patterns and lifestyle of a nation. Considering the high-income status of the selected economies, it is expected that they might rely on emerging economies to fulfill domestic demand through the acquisition of consumer products with high carbon intensity. Moreover, it can be argued that stringent environmental regulations have effectively optimized the positive aspects that resulted from environmentally friendly innovations, leading to lower emissions and enhanced efficiency. Consequently, it is logical to assert that the selected economies should certainly persist in their investment and promotion of green technology, which contributes to mitigating the impact of environmental degradation and contribute to an elevated quality of life. Consequently, it is rational to contend that the selected countries need to increase their investments and promote the adoption of environmentally friendly technology. These measures would alleviate the negative impact of environmental degradation and contribute to the creation of a sustainable and ecologically sound environment for the next generation. The current surge in emissions and the limited time-frame to achieve mitigation goals have exacerbated the urgency of this

battle. Enhancing energy saving plays a crucial role in combating global warming. Energy saving is essential for reducing global energy consumption and meeting the increasing demand for energy services. Implementing innovative environmentally-friendly policies could potentially reduce energy commodity prices. Subsequently, energy saving contributes to decreasing the energy consumed during the production of each unit, thereby enhancing the environment. Energy saving not only reduces manufacturing costs through innovation and infrastructure upgrades but also encourages consumers to save funds and decrease their ecological impact. This contributes to the worldwide effort to alleviate the impacts of climate change. Moreover, it is rational to assert that stringent environmental regulations have optimized the benefits derived from sustainable innovations, eventually leading to reduced emissions and enhanced efficiency. Therefore, it is logical to conclude that the selected countries should persist in investing in and endorsing the adoption of green technology, which would alleviate the impact of environmental degradation and contribute to an improved quality of life. Hence, it is logical that the G7 countries should invest more and promote the adoption of green technology, which will reduce the impact of environmental degradation and encourage to achieve the target of net zero emissions and contribute to the creation of a sustainable future for the next generation.

6.1. Policy implications

Although global efforts to address declining biodiversity and ecological degradation are in place, the degree at which these challenges continue to exacerbate remains elevated in all regions globally. Currently, there is a concerning decline in the state of the natural environment, frequently attributed to human activities. To effectively address the severe impacts of climate change, it is essential to take proactive measures to reduce the emissions of GHG that contribute to global warming. This study emphasizes the development of policies that can effectively contribute to the common objective of achieving net zero emissions by 2050, while also ensuring that the global temperature rise remains under the limit of 1.5 °C. The choice of these policies considers the distinct energy status, economic and social structures, and geographical factors of each country. (i) In order to effectively attain the objectives stipulated in the Paris Agreement and COP27, it is imperative to implement substantial innovative incentives, industrial strategies, public and private investments. To achieve these objectives, it is imperative to secure these investments in emerging industries, either nationally or globally. Further, fostering collaboration is essential to address the existing investment deficit in the transition towards clean energy, thereby reducing the global cost associated with this transition. (ii) The G7 economies, as founding members of the Climate Club, should prioritize the adoption of decarbonized industrial production as the prevalent business model in the long run. This strategy will not only promote green innovation and sustainable economic growth but also help tackle challenges such as carbon leakage and other potential obstacles to mitigation efforts, while simultaneously ensuring adherence to environmental regulations. (iii) The G7 nations should intensify their efforts to implement more rigorous environmental regulations that enable the transition towards a sustainable economy. This transition should be supported by collaborative commitments to adopt policies that effectively contribute to achieving a significant change towards cleaner economies, while also safeguarding the environment and reinforcing the rules-based global structure. It is imperative to maintain constant encouragement for the research, development, and implementation of green technologies, because these technologies play a crucial role in facilitating the expedient pursuit of environmental sustainability. (iv) The G7 nations persist in their collaborative efforts to promote the affordability, accessibility, and desirability of clean technology and sustainable solutions on a global scale. These attempts primarily focus on renewable energy technologies and environmental regulations and is carried out by various global efforts.

Moreover, it is imperative for G7 economies to actively promote the adoption of sustainable practices in developing and emerging economies. Simultaneously, there is a need to intensify collective attempts in order to achieve the objectives stated in the 2030 Agenda. The Clean Energy Economy Action Plan (EEAP) necessitates collaborative efforts to establish trade policies that reduce carbon emissions in economies, expedite the advancement of robust clean energy supply chains, foster the growth of mutually beneficial markets for clean energy goods and services, and facilitate substantial investments from public and private sectors in climate and energy security. These efforts attempt to address collective energy and climate challenges, thus ensuring a more stable and sustainable future. The study findings have several implications for emerging economies. The primary objective of sustainable development for developing countries should be the reduction of total emissions while maintaining significant levels of economic growth. These nations ought to seek a suitable equilibrium between fostering economic growth and safeguarding the integrity of their environment. An effective strategy for achieving their ultimate goal of environmental preservation entails implementing rigorous environmental regulations, promoting technology advancements, transitioning to a carbon-neutral economy, and enhancing energy efficiency. These nations require to establish incentives to promote the purchase of ecologically friendly products and services. Attracting FDI that promotes new INV and clean energy sources can further increase their efforts to produce sustainable energy and improve environmental quality.

6.2. Limitations of the study

This research is valuable for policymakers in G7 nations. Regulators and other stakeholders should emphasize FD, FDI, and FG, while simultaneously considering the deployment of clean and environmentally friendly technologies to achieve long-term economic objectives. The implementation of stringent environmental regulations pertains to the process through which G7 nations might attain their zero-carbon emission objectives and consequently preserve the environment. Prior to reaching a conclusion in this analysis, it is imperative to realize that no policy framework should be considered generally applicable, as it is highly difficult to incorporate all pertinent policy elements in the limitations of a distinct paradigm. This study also adheres to this approach. The present study examined the interaction of the environmental policy aspect between green technology innovation and environmental quality. Furthermore, this study also considered other economic aspects, such as economic growth, as well as financial elements including FD, FG, and FDI, since these factors play a crucial role in the attainment of carbon neutrality targets by employing DOLS, FMOLS, CCR and MMQR econometric approaches within the G7 economies. Future research in the developed world could incorporate additional economic factors such as trade and industrialization, social factors such as poverty, and financial factors such as green stocks, blue stocks, and remittances by employing other econometric techniques to provide valuable empirical evidences. These factors could provide valuable insights into the various components of environmental sustainability.

Declarations Ethics approval and consent to participate

Not Applicable.

Funding

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analysis, Data curation, Conceptualization. **Hu Xuhua:** Supervision, Methodology. **Hind Alnafisah:** Writing – review & editing, Funding acquisition. **Muhammad Rehan Akhtar:** Writing – original draft, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgement

This work is supported by “Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2024R549), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia”.

Consent for publication

Not Applicable

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